

Car Insurance and Pricing Estimation System

CENG 476

Ata Doruk Çakmak

202111030

Utku Bora Efil

202211023

Öykü Çoban

202111066

Date:

Abstract

This project presents a machine learning-based system for estimating car insurance costs and market prices. Users can input vehicle details—such as year, mileage, fuel type, and transmission—to receive on-demand predictions using trained Random Forest models. Developed with Python and Streamlit, the application incorporates data preprocessing, encoding of categorical variables, and interactive visualizations to support intuitive exploration and analysis. Although the insurance pricing logic is based on a realistic custom formula rather than real-world data, the system serves as a valuable tool for individuals and educational purposes. It demonstrates how machine learning can aid data-driven decision-making in vehicle valuation and insurance estimation.

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1. Introduction

Accurately estimating auto insurance premiums and market prices is a major concern for car owners, purchasers, and resellers. Conventional evaluation techniques frequently lack scalability, are manual, and are inconsistent. Machine learning provides a data-driven solution to this problem, producing predictions that are more automated and reliable.

This project introduces a web-based application that allows users to enter vehicle information and receive instant predictions for both insurance premiums and market prices. Built with Python and Streamlit, the system uses a trained Random Forest algorithm to make predictions based on key features such as year, mileage, fuel type, and transmission. Additionally, the application includes interactive data visualizations that help users explore how different factors affect pricing and risk.

While the insurance pricing logic is modeled using a custom risk factor formula rather than real-world insurance datasets, the system demonstrates the potential of combining machine learning with user-friendly interfaces to support vehicle evaluation and pricing education.

2. Technologies Used

Programming Language: Python

Web Framework: Streamlit

Data Handling and Visualization Libraries: pandas, NumPy (data processing) matplotlib, seaborn (data visualization)

Machine Learning: scikit-learn

Model used: RandomForestRegressor for both car price prediction and insurance estimation

Model Persistence: joblib (used to save and load trained models)

Plotly: For Interactive Visualizations

3. Project Structure

data/ - Contains the dataset files

src/

- data_processor.py: handles preprocessing and feature engineering
- model_trainer.py: trains and saves the Random Forest model
- utils.py: includes helper functions for risk calculation, formatting, and validation

app.py: Streamlit interface connecting all components

requirements.txt: Lists all project dependencies

.venv/ and venv/: Virtual environment directories for project isolation

4. Data Processing

The data processing pipeline includes several crucial steps:

- Handling missing or inconsistent values to ensure data quality
- Converting age ranges to numeric values by taking midpoints
- Mapping income categories to numeric values
- Converting categorical variables using label encoders
- Handling special cases for numeric columns (credit score, annual mileage, violations)
- Feature scaling where applicable to normalize numerical data ranges
- Splitting the dataset into training and testing subsets for model validation
- Basic exploratory data analysis (EDA) to understand distributions and correlations

5. Model Training

Machine learning models were trained to predict two primary targets: car insurance premiums and vehicle market prices. The training process involved:

- Using the preprocessed dataset to train regression models
- Splitting the data into training and testing sets (80-20 split)
- Applying Random Forest Regression with 100 estimators
- Evaluating models using metrics like Mean Squared Error (MSE) and R-squared (R^2) scores
- Implementing feature importance analysis
- Caching trained models for improved application performance

6. Application Interface

The user interface is developed using Streamlit, which provides a simple and interactive web platform. Key features include:

- Custom CSS styling for enhanced visual appeal and user experience
- Input widgets such as dropdown menus and sliders for car details
- Interactive visualizations using Plotly for better data exploration
- Caching mechanisms ([@st.cache_data](#) and `@st.cache_resource`) for improved performance
- Real-time predictions using trained models
- Similar profile finding functionality to compare with other users
- Responsive layout with wide-screen optimization

7. Project Optimizations

- Implementation of Streamlit caching for data loading and model training
- Efficient data preprocessing pipeline
- Optimized feature engineering for real-time predictions
- Responsive UI design with custom CSS
- Efficient handling of categorical variables

8. Conclusion

This project successfully implements a comprehensive car insurance and pricing prediction system that leverages machine learning and modern web technologies. The system demonstrates practical applications of data science in the automotive insurance domain through:

- A fully functional web application built with Streamlit that provides real-time predictions
- Robust data processing pipeline that handles various data types and edge cases
- Efficient machine learning implementation using Random Forest Regression
- Interactive visualizations that enhance user understanding of predictions
- Performance optimizations through caching and efficient data handling

The system serves as both a practical tool for users and an educational resource for understanding how machine learning can be applied to insurance and pricing estimation. While the insurance pricing logic is based on a custom formula rather than real-world data, the implementation provides a solid foundation that could be extended with real insurance data in the future.

The modular architecture and clean code structure make the system maintainable and extensible. Future enhancements could include:

- Integration with real-time market data APIs
- Additional machine learning models for comparison
- Enhanced visualization capabilities
- User authentication and profile management
- Historical prediction tracking and analysis

The project successfully demonstrates how modern web technologies and machine learning can be combined to create practical, user-friendly applications for complex estimation tasks.